AI System Design Framework

Product Requirements

Identify the business problem	Define the role of AI	Identify your constraints	Define success metrics

System Design

Identify data sources and retrieval techniques	Select system architecture and stack	Determine the UX and feedback loops

Evaluation and Monitoring

Define system guardrails	Define offline evaluation metrrics	Define online evaluation metrics

Production

Optimize for accuracy	Optimize for cost and latency	Optimize for reliability

Health Insurance Claims Review System

Background

Health insurers and health funds around the world employ medical reviewers to assess whether a requested treatment, procedure, or medication is covered under a patient's policy. This requires cross-referencing clinical documentation, coverage policies, clinical guidelines, and patient claims history. It is one of the most administratively intensive roles in healthcare operations.

What we are building

In this workshop, we are building an internal claims review system for a fictitious health insurance company called MDB Health. This system is used by medical reviewers to assess and adjudicate claims. The external-facing system through which healthcare providers submit claims and receive feedback on outcomes is out of scope.

Business Constraints

1. All denial decisions must include a human reviewer.
2. Reviewers must be able to override any system recommendation.
3. Complex cases, as defined by MDB Health's internal clinical guidelines, must be reviewed by a senior physician.
4. Patient data can only be processed within MDB Health's approved cloud environment.
5. Only models available on MDB Health's approved cloud provider are permitted.
6. All decisions must be auditable and traceable to a specific source document.

Performance Requirements

1. P95 response time for a recommendation must be under **10 seconds**.
2. System must handle up to **5,000 requests per day**.
3. Denial decisions must be communicated to the healthcare provider within **24 hours**.
4. Monthly model (embedding, LLM) costs must not exceed **$10,000**.
5. **99.9%** uptime SLA required

Exercise 1. Product requirements

Business Problem

Medical reviewers at MDB Health spend an average of **2 days** processing claim review requests — **4 times** the industry standard for non-urgent cases and **12 times** the industry standard for urgent ones. Delays at this scale postpone patient care and are a leading cause of reviewer burnout across the organization.

Define the role of AI

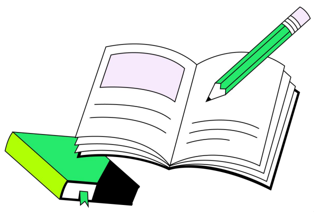
Role	Answer
Critical / Complementary (AI is the core / AI enhances the product)	
Reactive / Proactive (Triggered by user or event / Triggered by system)	
Level of AI autonomy	

Define success metrics

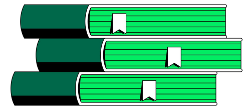
Instructions:

Define one SMART success metric for the MDB Health claims review system that reflects the core business problem.

S: Specific, **M**: Measurable, **A**: Achievable, **R**: Relevant, **T**: Timebound



Exercise 2. Data and retrieval



Instructions:

The data sources that are available for you to use to build the system are listed below. Identify the update frequency, processing needs and retrieval techniques for each of the data sources.

Data source	Where does it live?	Format
Clinical guidelines	Confluence	PDF
Internal coverage policies	Confluence	PDF
Patient claims history	MongoDB	BSON

💡 Discuss what additional data sources might improve the system's recommendations.

Estimate data update frequency

Instructions:

Estimate the update frequency of the data sources. Here are some options: Real-time, hourly, daily, monthly, quarterly, yearly, on-demand.

Data source	Update frequency
Clinical guidelines	
Internal coverage policies	
Patient claims history	

Exercise 2. Data and retrieval

Determine data processing needs

Instructions:

Determine the data processing needs of the data sources. Write down whether the data source needs **ingestion-time** or **inference-time** data processing, and a short note on the type of processing.

Recap:

Ingestion time: Done before ingesting data into your primary and/or final storage destination. Useful when data is unstructured or semi-structured (e.g., PDFs, Markdown), consists of long documents, contains PII that must be stripped before indexing, or doesn't match your retrieval format.

Inference time: Done before passing data to an LLM. Useful when data contains PII that must not reach the LLM, or doesn't match the LLM's input format.

Data source	Data processing
Clinical guidelines	
Internal coverage policies	
Patient claims history	

Exercise 2. Data and retrieval

Determine retrieval techniques

Instructions:

Determine the retrieval technique that best suits each data source. Choose one among: vector search, exact match, full-text search, hybrid search, or something else. If your response is “something else”, note down what that looks like.

Recap:

Vector search: Retrieves data based on similarity in meaning (semantic similarity).

Exact match: Retrieves data that matches specific criteria.

Full-text search: Retrieves data based on keyword matches.

Hybrid search: Combine the results of different search techniques.

Data source	Retrieval technique
Clinical guidelines	
Internal coverage policies	
Patient claims history	

Exercise 3. System architecture

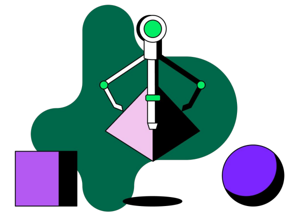
Trace a claim from submission to decision

Instructions:

A claim has been received by MDB Health's claims review system. In 5-7 steps, describe how the claim is reviewed and adjudicated. The outcome should be an approval or rejection.

1. Claim request and clinical notes are received by the claims review system.

Exercise 3. System architecture



Identify the design patterns

Instructions:

Based on the the system flow, what design patterns would you use to implement this system?

Recap:

RAG: Tasks that require information that might not be present in the LLM's pre-trained knowledge.

Agents: Complex tasks that don't have a structured workflow.

Controlled flow: LLMs perform tasks within structured workflows.

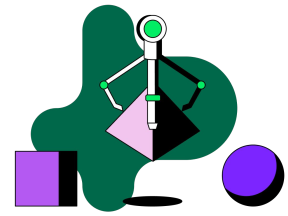
LLM-as-a-router: LLMs categorize and route requests to downstream workflows.

Human-in-the-loop: Tasks requiring human intervention.

Fine-tuning: Fix consistent behavioral model failures, high-lift.

Design pattern	Why?

Exercise 3. System architecture



Define UX and feedback mechanisms

Instructions:

Think about the interface between the MDB Health claims review system and its users. Based on all the information you have so far, answer the following questions.

Remember, this system is for MDB Health medical reviewers, not the healthcare providers submitting claims or the patients whose care is being reviewed.

Decision	Answer
What does the system take as input?	
What does the system produce as output?	
Where does the system live?	
What triggers the system?	
What is the human's role?	
How does the system explain itself?	
How do users give feedback?	

Exercise 4. Evaluation and monitoring

Design system guardrails

Instructions:

Define what are acceptable inputs and outputs to the system. Come up with **one** input and output guardrail each.

Type	Guardrail
Input	
Output	

Define offline evaluation metrics

Instructions:

Define **one** system metric per category that you would use to evaluate the system before deploying it to production.

Category	Metric
Input guardrail compliance	
Output guardrail compliance	
Response quality	
Domain-specific accuracy	
Performance	

Exercise 4. Evaluation and monitoring

Define online evaluation metrics

Instructions:

Define **one** user feedback and behavioral metric you would use to monitor the system in production.

Category	Metric
User feedback (Explicit feedback signals)	
Behavioral (User actions that indicate implicit feedback)	

SOLUTION

Exercise 1. Product requirements

Business Problem

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Define the role of AI

Role	Answer
Critical / Complementary (AI is the core / AI enhances the product)	Complementary
Reactive / Proactive (Triggered by user or event / Triggered by system)	Reactive
Level of AI autonomy	Semi-autonomous

Define success metrics

Instructions:

Define one SMART success metric for the MDB Health claims review system that reflects the core business problem.

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Reduce the average processing time for urgent claim review requests from 2 days to 4 hours within 90 days of launch.

SOLUTION

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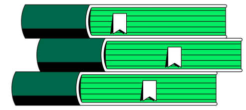

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Estimate data update frequency

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Data source	Update frequency
Clinical guidelines	On-demand
Internal coverage policies	On-demand
Patient claims history	Real-time

SOLUTION

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Data source	Data processing
Clinical guidelines	Ingestion-time
Internal coverage policies	Ingestion-time
Patient claims history	None

SOLUTION

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Hybrid search: Combine the results of different search techniques.

Data source	Retrieval technique
Clinical guidelines	Hybrid search
Internal coverage policies	Hybrid search
Patient claims history	Exact match

SOLUTION

Exercise 3. System architecture

Trace a claim from submission to decision

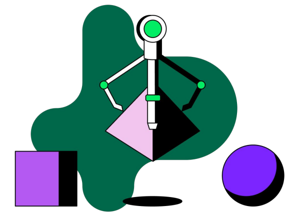
Instructions:

A claim has been received by MDB Health's claims review system. In 5-7 steps, describe how the claim is reviewed and adjudicated. The outcome should be an approval or rejection.

1. Claim request and clinical notes are received by the claims review system.
2. Patient's claims history, and relevant clinical guidelines and coverage policies are retrieved.
3. An LLM generates a coverage recommendation.
4. If it is a complex case, escalate the recommendation to a senior physician.
5. If the recommendation is a denial, escalate it to a medical reviewer.
6. Log the final decision along with the reasoning.

SOLUTION

Exercise 3. System architecture



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LLM-as-a-router: LLMs categorize and route requests to downstream workflows.

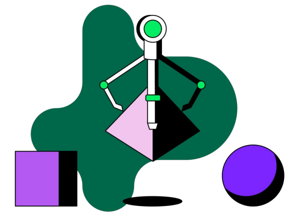
Human-in-the-loop: Tasks requiring human intervention.

Fine-tuning: Fix consistent behavioral model failures, high-lift.

Design pattern	Why?
RAG	Retrieve clinical guidelines, coverage policies and past claim history to inform the decision.
Controlled flow	LLMs perform tasks within structured workflows.
Human-in-the-loop	Denial decisions and complex cases need to be reviewed by a human.

SOLUTION

Exercise 3. System architecture



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Instructions:

Think about the interface between the MDB Health claims review system and its users. Based on all the information you have so far, answer the following questions.

Remember, this system is for MDB Health medical reviewers, not the healthcare providers submitting claims or the patients whose care is being reviewed.

Decision	Answer
What does the system take as input?	Claim request form, clinical notes
What does the system produce as output?	Approval or denial verdict along with an explanation
Where does the system live?	Embedded within MDB Health's existing claims management system
What triggers the system?	A claim submission
What is the human's role?	Review the system's assessment and make the final decision
How does the system explain itself?	By citing the clinical guidelines and coverage policies that informed the assessment
How do users give feedback?	By overriding the verdict and logging a reason, flagging irrelevant citations

SOLUTION

Exercise 4. Evaluation and monitoring

Design system guardrails

Instructions:

Define what are acceptable inputs and outputs to the system. Come up with **one** input and output guardrail each.

Type	Guardrail
Input	Claims should have supporting clinical documentation
Output	Final result should provide citations

Define offline evaluation metrics

Instructions:

Define **one** system metric per category that you would use to evaluate the system before deploying it to production.

Category	Metric
Input guardrail compliance	Claim rejection rate
Output guardrail compliance	Missing citation rate
Response quality	Faithfulness
Domain-specific accuracy	Override rate
Performance	Average processing time

SOLUTION

Exercise 4. Evaluation and monitoring

Define online evaluation metrics

Instructions:

Define **one** user feedback and behavioral metric you would use to monitor the system in production.

Category	**Metric**
User feedback (Explicit feedback signals)	# of incorrect citations flagged
Behavioral (User actions that indicate implicit feedback)	Average review time